

1.1 -> done

1.2

gcc -E hello.c -o hello.i result:

```
GNU nano 7.2
# 0 "hello.c"
# 0 "<built-in>"
# 0 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 0 "<command-line>" 2
# 1 "hello.c"
# 1 "/usr/include/stdio.h" 1 3 4
# 28 "/usr/include/stdio.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 1 3 4
# 33 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 3 4
# 1 "/usr/include/features.h" 1 3 4
# 394 "/usr/include/features.h" 3 4
# 1 "/usr/include/features-time64.h" 1 3 4
# 20 "/usr/include/features-time64.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 21 "/usr/include/features-time64.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 1 3 4
# 19 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 20 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 2 3 4
# 22 "/usr/include/features-time64.h" 2 3 4
# 395 "/usr/include/features.h" 2 3 4
# 502 "/usr/include/features.h" 3 4
```

gcc -S hello.c -o hello.s result:

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -S hello.c -o hello.s
hello.c: In function 'main':
hello.c:5:9: warning: implicit declaration of function 'printf' [-Wimplicit-function-declaration]
    5 |         printf("%s\n", MESSAGE);
      |         ~~~~~
hello.c:1:1: note: include '<stdio.h>' or provide a declaration of 'printf'
+++ |+#include <stdio.h>
    1 |
hello.c:5:9: warning: incompatible implicit declaration of built-in function 'printf' [-Wbuiltin-declaration-mismatch]
    5 |         printf("%s\n", MESSAGE);
      |         ~~~~~
hello.c:5:9: note: include '<stdio.h>' or provide a declaration of 'printf'
```

```
.file "hello.c"
.text
.section .rodata

.LC0:
.string "Bonjour Ynov!"
.text
.globl main
.type main, @function

main:
.LFB0:

.cfi_startproc
endbr64
pushq %rbp
.cfi_def_cfa_offset 16
.cfi_offset 6, -16
movq %rsp, %rbp
.cfi_def_cfa_register 6
leaq .LC0(%rip), %rax
movq %rax, %rdi
call puts@PLT
movl $0, %eax
popq %rbp
.cfi_def_cfa 7, 8
```

gcc -S hello.c -o hello.o result

[illegible]

```
gcc hello.c -o hello result:
```

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ ./hello
Bonjour Ynov!
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$
```

File cmd on each file result:

```
hello hello.c hello.i hello.o hello.s
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ file hello
hello: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=d2215bde6f814bf883959246380abb93cd513cff, for GNU/Linux 3.2.0, not stripped
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ file hello.c
hello.c: C source, ASCII text
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ file hello.i
hello.i: C source, ASCII text
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ file hello.o
hello.o: ELF 64-bit LSB relocatable, x86-64, version 1 (SYSV), not stripped
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ file hello.s
hello.s: assembler source, ASCII text
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$
```

Questions :

1. The file now includes all the parameters structures and methods that the `#include` parameter needs, and all the dependencies of `stdio.h`. The `MESSAGE` got replaced with the

screenshot below.

```
# 4 "hello.c"
int main(void) {
    printf("%s\n", "Bonjour Ynov!");
    return 0;
}
```

820 Lines of code :

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ wc -l hello.i
820 hello.i
```

2. The Code inside hello.s is assembly code. Basically the last human understandable code before it becomes machine code

3. Already done above in a screenshot but the difference is that hello.o is an object file and hello is an executable

4. gcc hello.c -o hello permits doing all of this in one command and -Wall -Wextra for error management

1.3

Directive	Role	Exemple
<code>#include <stdio.h></code>	Insere le contenu du fichier header	Fonctions standard
<code>#include "mon.h"</code>	Idem, cherche d'abord dans le dossier courant	Headers perso
<code>#define PI 3.14159</code>	Definit une constante (remplacement textuel)	<code>PI</code> → <code>3.14159</code>
<code>#define MAX(a,b) ((a)>(b)? (a):(b))</code>	Macro avec parametres	Substitution
<code>#ifdef DEBUG / #endif</code>	Compilation conditionnelle	Mode debug

1.4

Quand votre code appelle `printf`, le compilateur ne connaît pas son adresse en mémoire. C'est le **linker** qui relie votre code aux bibliothèques (comme la libc).

Erreur fréquente : `undefined reference to 'fonction'` signifie que le linker n'a pas trouvé l'implémentation. Causes : oubli d'un fichier `.o`, oubli de `-lm` pour les fonctions math.

2.1

Option	Role	Exemple
<code>-o nom</code>	Definit le nom du fichier produit	<code>gcc main.c -o app</code>
<code>-c</code>	Compile sans lier (produit un .o)	<code>gcc -c main.c</code>
<code>-Wall</code>	Active tous les warnings courants	<code>gcc -Wall main.c</code>
<code>-Wextra</code>	Warnings supplémentaires	<code>gcc -Wall -Wextra main.c</code>
<code>-g</code>	Inclut les infos de débogage (pour gdb)	<code>gcc -g main.c -o app</code>
<code>-O2</code>	Optimisations de production	<code>gcc -O2 main.c -o app</code>
<code>-std=c11</code>	Choisit la version du standard C	<code>gcc -std=c11 main.c</code>

Conseil pédagogique : Pendant tout le module, compilez TOUJOURS avec `-Wall -Wextra`. Les warnings vous évitent des heures de debug.

2.2

Compilation Step :

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc test_options.c -o test
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall test_options.c -o test
test_options.c: In function 'main':
test_options.c:3:9: warning: unused variable 'x' [-Wunused-variable]
  3 |     int x;
    |     ^
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall -Wextra test_options.c -o test
test_options.c: In function 'main':
test_options.c:3:9: warning: unused variable 'x' [-Wunused-variable]
  3 |     int x;
    |     ^
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$
```

Questions :

1. We obtain a warning telling us that a variable named x was not used
2. Inside of the code that we have we need to use variable x somewhere or completely delete it
3. It's important to handle warnings even if the program compiles because it could lead to issues down the road. Fixing any problems now will make sure that your code is 100% ready

3.1

```
#include <stdio.h>

int main(void) {
    int    age      = 19;      // entier signe (32 bits)
    float  taille    = 1.78f;  // flottant simple precision
    double pi        = 3.14159; // flottant double precision
    char   initiale  = 'A';    // un seul caractere (8 bits)

    printf("Age=%d, Taille=%.2f, Init=%c\n", age, taille, initiale);
    return 0;
}
```

3.2

Type	Taille	Plage approximative	Format printf
<code>char</code>	1 octet	-128 a 127	<code>%c</code> (caractere) ou <code>%d</code> (numerique)
<code>unsigned char</code>	1 octet	0 a 255	<code>%u</code>
<code>short</code>	2 octets	-32 768 a 32 767	<code>%hd</code>
<code>int</code>	4 octets	-2.1 milliards a +2.1 milliards	<code>%d</code>
<code>unsigned int</code>	4 octets	0 a 4.29 milliards	<code>%u</code>
<code>long</code>	8 octets	tres grand	<code>%ld</code>
<code>float</code>	4 octets	~6 chiffres significatifs	<code>%f</code>
<code>double</code>	8 octets	~15 chiffres significatifs	<code>%lf</code> ou <code>%f</code>

Astuce : Utilisez `sizeof(int)` dans votre code pour obtenir la taille reelle sur votre systeme.

3.3

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ ./types
=== Tailles des types C ===
char   : 1 octet(s)
short  : 2 octet(s)
int    : 4 octet(s)
long   : 8 octet(s)
float  : 4 octet(s)
double : 8 octet(s)
```

For the question yes they match exactly the table contents above. As to why a machine would

say differently to another it could be a memory thing ? meaning one machine uses different memory allocation depending on what type of compiler and the architecture it uses ?

3.4 Already acquired

3.5

Result

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall -Wextra calculs.c -o calculs
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ ./calculs
sum:22
sub:12
mult:85
div:3
mod:2
float:3.40
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$
```

Code

```
#include <stdio.h>

int main(void) {
    int a = 17;
    int b = 5;
    int addition = a + b;
    int subtraction = a - b;
    int multiplication = a * b;
    int division = a / b;
    int mod = a % b;
    printf("sum:%d\n sub:%d\n mult:%d\n div:%d\n mod:%d\n", addition, subtraction, multiplication, division, mod);
    printf("float:%.2f\n", (float)a / b);
    return 0;
}
```

4.1

```

#include <stdio.h>

int main(void) {
    int    age;
    float  taille;
    char   prenom[30];

    printf("Quel est ton prenom ? ");
    scanf("%29s", prenom);

    printf("Quel age as-tu ? ");
    scanf("%d", &age);

    printf("Quelle taille (en m) ? ");
    scanf("%f", &taille);

    printf("Bonjour %s, %d ans, %.2f m.\n", prenom, age, taille);
    return 0;
}

```

Points importants :

- `scanf` attend l'**adresse memoire** d'une variable : on ecrit `&age` (sauf pour les chaines, car un nom de tableau est deja une adresse)
- Le `%29s` limite l'entree a 29 caracteres pour eviter le debordement (la chaine fait 30 cases)
- Sans le `\n` dans printf, le curseur reste sur la meme ligne -- pratique pour les prompts

4.2

Test 1 :

we have warnings here because apparently scanf does not support %.2f only printf does

```

ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall -Wextra fiche_interactive.c -o fiche_interactive
fiche_interactive.c: In function 'main':
fiche_interactive.c:15:17: warning: unknown conversion type character '.' in format [-Wformat=]
   15 |         scanf("%.2f", &grade);
      |         ^
fiche_interactive.c:15:15: warning: too many arguments for format [-Wformat-extra-args]
   15 |         scanf("%.2f", &grade);
      |         ^~~~~~
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ ./fiche_interactive
What is your name ?
Ayoub
How old are you ?
21
What's your average grade ? (floatpoint)
19.75
Hello Ayoub, you're 21 years old and your average grade of 0.00 will be used for ai training ! be happy before the end !
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$

```

Test 2 :

Funnily enough the conversion from %f to %.2f does work ! :D

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall -Wextra fiche_interactive.c -o fiche_interactive
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ ./fiche_interactive
What is your name ?
Luka
How old are you ?
19
What's your average grade ? (floatpoint)
16.48
Hello Luka, you're 19 years old and your average grade of 16.48 will be used for ai training ! be happy before the end !
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$
```

Code :

```
#include <stdio.h>

int main(void) {
    int age;
    float grade;
    char name[10];

    printf("What is your name ?\n");
    scanf("%10s", name);

    printf("How old are you ?\n");
    scanf("%d", &age);

    printf("What's your average grade ? (floatpoint)\n");
    scanf("%f", &grade);

    printf("Hello %s, you're %d years old and your average grade of %.2f will be used for ai training ! be happy before the end !\n", name);
    return 0;
}
```

5.1

Erreurs (errors)	Avertissements (warnings)
Le code ne compile PAS	Le code compile, mais sent mauvais
expected ';' before return	unused variable 'x'
undefined reference to 'printf'	implicit declaration of function
A resoudre AVANT de pouvoir lancer	A traiter serieusement

5.2

Missing ; :

It's self explanatory it's telling us exactly where we are missing and what we are missing, also another error i unintentionally created is the array size error in name[] and also an undeclared

error in age XDDD

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall -Wextra erreurs.c -o erreurs
erreurs.c: In function 'main':
erreurs.c:4:16: error: expected ';' before 'int'
   4 |         int age
     |                ^
     |                ;
   5 |         int name[];
     |         ~~~
erreurs.c:5:13: error: array size missing in 'name'
   5 |         int name[];
     |         ~~~~~
erreurs.c:6:25: error: 'age' undeclared (first use in this function)
   6 |         printf("%d %s", age, name);
     |                        ^~~
erreurs.c:6:25: note: each undeclared identifier is reported only once for each function it appears in
```

Code :

```
#include <stdio.h>

int main(void) {
    int age;
    int name[];
    printf("%d %s", age, name);
}
```

Incompatible type :

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall -Wextra erreurs.c -o erreurs
erreurs.c: In function 'main':
erreurs.c:6:25: error: incompatible types when initializing type 'float' using type 'char *'
   6 |         float garbage = "bottles";
     |                        ~~~~~
erreurs.c:6:15: warning: unused variable 'garbage' [-Wunused-variable]
   6 |         float garbage = "bottles";
     |         ~~~~~
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$
```

Code :

```
#include <stdio.h>
```

```
int main(void) {
    int age = 20;
    char name[19];
    float garbage = "bottles";
    printf("%d %s", age, name);
}
```

Non-declared function :

Actual error we are looking for is at the end of the screenshot

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall -Wextra erreurs.c -o erreurs
erreurs.c: In function 'main':
erreurs.c:5:13: error: array size missing in 'name'
   5 |         int name[];
     |             ^~~~
erreurs.c:6:21: warning: format '%s' expects argument of type 'char *', but argument 3 has type 'int *' [-Wformat=]
   6 |         printf("%d %s", age, name);
     |                ^~   ^~~~~
     |                |     |
     |                char * int *
     |                %ls
erreurs.c:8:9: warning: implicit declaration of function 'maFonction' [-Wimplicit-function-declaration]
   8 |         maFonction();
     |         ^~~~~~
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$
```

Code :

```

GNU nano 7.2
#include <stdio.h>

int main(void) {
    int age;
    int name[];
    printf("%d %s", age, name);

    maFonction();

    return 0;
}

```

6.0

Compilation and run :

```

ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ nano convertisseur.c
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ gcc -Wall -Wextra convertisseur.c -o convertisseur
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ ./convertisseur
=== Universal Converter ===
1. Km -> Mile
2. Kg -> Pound
3. C -> F
4. Quit
Your Choice: 1
Value to Convert: 20
Result: 12.43

```

Invalid type (just return 0):

```

ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ ./convertisseur
=== Universal Converter ===
1. Km -> Mile
2. Kg -> Pound
3. C -> F
4. Quit
Your Choice: 1
Value to Convert: urdad
Result: 0.00
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$

```

Invalid Choice :

```
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$ ./convertisseur
=== Universal Converter ===
1. Km -> Mile
2. Kg -> Pound
3. C -> F
4. Quit
Your Choice: 5
Choose a correct value ya Nonce
ashen@Ashen:/mnt/c/Users/dznel/Desktop/Ynov/compiledLanguages/Day 2$
```

Code (calc part) :

```
if (choice == 4)
{
    printf("Buh Bye\n");
    return 0;
}

if (choice == 1 || choice == 2 || choice == 3)
{
    printf("Value to Convert: ");
    scanf("%lf", &value);

    if (choice == 1)
        result = value * 0.621371;
    else if (choice == 2)
        result = value * 2.20462;
    else
        result = value * 9.0 / 5.0 + 32.0;

    printf("Result: %.2f\n", result);
}
else
{
```